

Shazia Ruybal-Pesántez

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My expertise lies at the intersection of infectious disease epidemiology, modeling, and pathogen surveillance, with over 10 years research experience in applied ('field'), genomic, and sero-epidemiology. My research integrates bioinformatics, population genetics, computational epidemiology, statistical and mathematical modelling, and advanced analytics to better understand infection dynamics, particularly malaria and other vector-borne diseases. My work has leveraged these insights to support disease control and elimination efforts as well as outbreak response and pandemic preparedness. I have long-standing collaborations spanning the Americas, West and East Africa, and Asia-Pacific, with strong links to disease control programs and a focus on capacity-building, equitable partnerships, and enhancing surveillance initiatives.

Current Appointments

Barcelona Institute for Global Health (ISGlobal)

Junior Research Leader

Barcelona, Spain

Nov 2025–Present

MRC Centre for Global Infectious Disease Analysis, Imperial College London

Visiting Researcher, Department of Infectious Disease Epidemiology

London, UK

Feb 2025–Present

Universidad San Francisco de Quito

Adjunct Associate Professor, Institute of Microbiology, College of Biological and Environmental Sciences

Quito, Ecuador

Mar 2020–Present

Education

The University of Melbourne

Ph.D. Genetic Epidemiology

Melbourne, Australia

2018

- Recipient of the Melbourne International Engagement Award and Melbourne International Fee Remission Scholarship (2014-2018) awarded to a limited number of international students based on academic merit to cover full tuition and living costs
- **Thesis:** Genetic epidemiology of the *Plasmodium falciparum* reservoir of infection in Bongo District, Ghana [↗](#)

The Colorado College

B.A. Biology

Colorado, USA

2012

- Recipient of the Presidential Scholarship (2008-2012) awarded to students with outstanding academic and extracurricular achievement
- Recipient of the James Wilkes Memorial Prize in Biology (2011) awarded to the most outstanding undergraduate minority student in Biology
- Recipient of the Venture Grant (2012) awarded to students with academic merit to support innovative project proposals

Maastricht University

Study Abroad Program (Spring 2011)

Maastricht, Netherlands

2011

- Undertook courses in immunology, human physiology and public health policymaking

Scientific Career Overview

Evidence of Esteem

- 32 peer-reviewed publications, 8 first/equal-contribution or senior author and 4 pre-prints (incl Trends in Parasitology, PLOS Genetics, iScience, Nature Comms, PLOS Biology)
- 1 book chapter ("Partnership-based approach to infectious disease research in Papua New Guinea")
- Academic Editor for PLOS Global Public Health, Guest Editor for Frontiers in Genetics Special Issue, and active peer reviewer (incl Nature Comms, PLOS Genetics, JID, Bioinformatics)
- Track record of securing funding and recognition of my work, including 1 independent fellowship, 9 grants (3 as PI; 6 as Co-I) and 16 awards
- 19 invited talks (incl ASTMH, AMMnet, Center for Research on Health in Latin America, UNC-Chapel Hill, LSHTM, Zanzibar Malaria Elimination Programme)
- 26 conference presentations (10 oral and 17 poster, 6 travel awards, incl at ASTMH, Wellcome GEM, MAM Australia, user!)
- 5 conference committees and 14 workshop organization (8 as lead organizer)
- Formal supervision of 3 PhD (2 to completion; 1 ongoing), 2 Masters (1 to completion; 1 ongoing), 2 short-term undergraduate and 1 Honour students (to completion). Supervision/mentorship of field research/technical staff (4 staff in PNG, now 1 undertaking MSc; 3 staff in Ecuador, now 1 undertaking PhD; 1 RA in Australia). Serving on 2 PhD committees at University of Melbourne.
- 6 consortia memberships (incl Applied Modeling for Malaria Network, PlasmogenEpi network, Australian Centres of Excellence – Malaria Elimination, NTDs Elimination)
- President of Ecuadorian Research and Entrepreneurship Network (2015-2017) and R-Ladies Melbourne (2019-2022)

Professional Experience

Barcelona Institute for Global Health (ISGlobal)

Junior Research Leader

Barcelona, Spain

Nov 2025–Present

- Leading an independent research program integrating genomic and computational epidemiology, modelling, and molecular surveillance to investigate vector-borne disease transmission dynamics, particularly malaria outbreaks, resurgence and reemergence in various settings. Funded through a la Caixa Foundation Junior Leader Fellowship.
- Developing novel statistical and modeling frameworks that combine pathogen genetic and epidemiological data for actionable surveillance and outbreak analytics
- Partnering with key international collaborators and disease control program partners to translate genomic insights into evidence-based elimination strategies and open-source tools

Menzies School of Health Research

Consultant Genomic Epidemiologist and Modeller

Consultant (remote)

Aug 2025–Nov 2025

- Apply population genetics and statistical modelling to analyse *P. vivax* genomic epidemiological data from outbreak response efforts on the Thai-Myanmar border and China-Myanmar border aiming to understand imported malaria transmission dynamics
- Apply and extend statistical models to understand *P. vivax* recurrent infection dynamics in various settings

University of North Carolina at Chapel Hill

Consultant Genomic Epidemiologist and Modeller

Consultant (remote)

Feb 2025–Jul 2025

- Develop novel methods and computational tools to analyse genomic data to support outbreak response and transmission chain inference
- Apply new analytical approaches (statistical modelling and network machine learning approaches) to analyse genomic epidemiological data from a longitudinal study in Tanzania aiming to understand imported malaria transmission dynamics
- Work closely with the Zanzibar Malaria Elimination Program in Tanzania using genomics to inform decision-making

Applied Epi

Instructor

Consultant (remote)

Jul 2023–Present

- Provide training courses (English and Spanish) in R for data analysis, visualization and advanced topics (e.g. GIS, stats, relational databases, dashboards, automated reports, Shiny apps) to applied epidemiologists and public health officials
- Translate and adapt training materials from English to Spanish
- Develop course curriculums and training materials for advanced topics (eg Shiny apps)

MRC Centre for Global Infectious Disease Analysis, Imperial College London

Research Associate, Department of Infectious Disease Epidemiology

London, UK

Oct 2022–Jan 2025

- My postdoc work involved the development of novel simulation-based approaches and analytical methods to generate evidence on the utility of genetic data for national malaria control and elimination programs, as well as computational tools to support study design and powering malaria molecular surveillance studies.
- Develop the SIMPLEGEN  simulation-based pipeline that combines mathematical models of malaria transmission with genetic models that can be used to systematically explore the utility of genetic data under different epidemiological conditions. This modelling tool will allow users to explore the "power" of genomic sampling designs in silico to help inform malaria genomic surveillance best practices.
- Develop the DRpower  Shiny app in collaboration with the WHO Global Malaria Programme to aid study design for monitoring *pfhrp2/3* gene deletion prevalence. This decision-support tool is now integrated in the updated WHO Master protocol .
- Support systematic reviews of existing mathematical models and epidemiological parameters for the nine WHO blue-print priority pathogens with epidemic potential and no approved vaccine/treatment and contribute to the development of the epireview  R package as a member of the Pathogen Epidemiology Review Group (MRC GIDA WHO Collaborating Centre for Infectious Disease Modelling)
- Outputs: First author publications (2), Co-author publications (11), Student supervision (3), Software (3)

Universidad San Francisco de Quito

Visiting Scientist, Institute of Microbiology

Quito, Ecuador

Oct 2021–Feb 2022

- Support the coordination and data analysis (ongoing) of a population-based cohort study in Quito to understand vaccine and infection-induced immunity to SARS-CoV-2 (ongoing). Supervised team of four research and clinical staff (now 1 as PhD student)
- Support data analyses on the genomic epidemiology and phylogenetics of different SARS-CoV-2 variants circulating in Ecuador (ongoing)
- Outputs: Student/staff supervision (4)

Ministry of Foreign Affairs & Migration/National COVID- 19 Emergency Response Committee

Consultant Epidemiologist

Consultant

Mar 2020–Dec 2020

- Provided internal situation reports on case/testing trends and epidemiological modelling (importation dynamics, Reff) to inform emergency planning

Ecuadorian Red Cross National Blood Bank

Consultant Epidemiologist

Consultant

Mar 2020–Dec 2020

- Led the establishment of SARS-CoV-2 sero-surveillance as part of the COVID-19 emergency response, delivered a real-time surveillance flexdashboard

Walter and Eliza Hall Institute of Medical Research

Melbourne, Australia

Research Officer, Population Health and Immunity Division

Feb 2019–Sep 2022

- My postdoc work involved the application of a suite of genomic epidemiology approaches to better understand residual and resurgent malaria transmission dynamics in the Asia-Pacific and Americas regions
- Supported the field implementation and lead the overall analysis of a 12-month longitudinal cohort study in Papua New Guinea as part of the Asia-Pacific ICEMR. The aim of this study is to understand the spatiotemporal risk factors for malaria infections (Pf and Pv) in 1000 individuals of all ages residing across four villages on the North Coast of PNG.
- Developed flexdashboards for real-time visualization of operational logistics and epidemiological trends, integrating study data with health facility surveillance data
- Managed, curated, and analyzed large epidemiological datasets from malaria, NTDs, and COVID-19 studies, integrating genomic and serological datasets using quantitative methods and reproducible workflows
- Supervised the development of an amplicon sequencing microhaplotype genotyping approach for *P. vivax* (PvAmpSeq) and analysis of genotyped samples from a clinical trial in Solomon Islands (ACT-Radical study) to discriminate relapses, recrudescence and reinfections.
- Outputs: Senior author publications (1), Co-author publications (6), Software (1), Honors and awards/grants (9), Student/staff supervision (7)

Burnet Institute

Melbourne, Australia

Honorary Research Officer, Vector-Borne Diseases and Tropical Public Health Working Group

Feb 2019–Sep 2022

- Supported field implementation, data/sample collection procedures, and analysis of cross-sectional surveys of NTDs and VPDs in PNG
- Supported data collection and analysis of the STRIVE PNG real-time VBD sentinel surveillance and response system, including quantitative methods and tools for multi-pathogen serological surveillance
- Outputs: Software (1), Book chapter (1)

The University of Melbourne

Melbourne, Australia

Research Fellow in Malaria Population Genetics

May 2018–Feb 2019

- Applied genomic epidemiology approaches that employed bioinformatic and population genetic methods to better understand the diversity and geographic population structure of *var* genes in Ecuador and Ghana
- Outputs: First author publications (1), Co-author publications (4), Student supervision (1)

Center for Research on Health in Latin America

Quito, Ecuador

Visiting Scientist

Nov 2017–Feb 2018

- I was awarded a JD Smyth Postgraduate Student Travel Award by the Australian Society for Parasitology to support a Researcher Exchange to establish a multi-institutional research collaboration
- Applied amplicon sequencing approaches ('varcoding') and population genetic analytical methods to examine resurgent malaria transmission patterns in pre-elimination settings of Ecuador
- Trained laboratory staff on *Plasmodium* genotyping protocols and delivered analytical skills workshops

The University of Melbourne

Melbourne, Australia

Postgraduate Researcher

Feb 2014–May 2018

- Developed a high-throughput amplicon sequencing genotyping tool ('varcoding') and customised computational and analytical methods for characterizing antigenic diversity in *P. falciparum*
- My PhD work involved the generation of microsatellite genotyping and *var* gene illumina sequence data ('varcoding') and downstream analysis using population genetic, bioinformatic, network and epidemiology approaches to relate parasite diversity data to epidemiological data
- Visited our field site in Bongo District, Ghana in 2014 and was directly involved in the field coordination and data collection as part of a cross-sectional survey in Bongo
- Outputs from my PhD: First-author publications (4), Co-author publications (5), Honors and awards (7), Student supervision (2)

New York University of School of Medicine

New York, USA

Research Assistant, Division of Medical Parasitology

Jun 2012–Jan 2014

- Analyzed the molecular epidemiology and population genetics of the *P. falciparum* *var* multi-gene family encoding the major variant surface antigen of malaria

New York Blood Center Lindsley F. Kimball Research Institute

New York, USA

Research Intern, Molecular Parasitology

Jul 2011–Aug 2011

- Research project: Polymorphisms in invasion ligand genes from Peruvian, Colombian and Brazilian *Plasmodium falciparum* field isolates
- Identified novel polymorphisms in DNA sequences coding for proteins critical to the invasion process and pathogenicity of *P. falciparum*

Supervision Experience

In addition to the formal supervision roles listed below, I have been actively involved in the supervision, mentorship and capacity-building of researcher officers and technical staff in partner institutions in Indonesia, Papua New Guinea and Ecuador. I currently serve on 2 PhD committees at University of Melbourne.

MMSc Student, Elma Nate

University of Papua New Guinea

Madang, Papua New Guinea

Aug 2023–Present

- Co-supervise a Medical Sciences Masters student on neglected tropical diseases molecular and sero-epidemiology
- Thesis: Investigating the burden of lymphatic filariasis and soil transmitted helminths infections in malaria endemic communities on the North Coast of Madang; Papua New Guinea

PhD Student Co-Supervisor, Belén Prado-Vivar

Walter and Eliza Hall Institute of Medical Research

Melbourne, Australia

Mar 2023–Present

- Co-supervise a Medical Biology PhD student on SARS-CoV-2 genomics and neglected tropical diseases sero-epidemiology
- Thesis: Understanding Neglected Tropical Diseases and Viral Infection dynamics in an Ecuadorian population cohort

Honours Student Co-Supervisor, Natalie Cerovac

Walter and Eliza Hall Institute of Medical Research

Melbourne, Australia

Feb–Dec 2021

- Designed and co-supervised an Honours research project in seroepidemiology for a Biomedical undergraduate student
- Thesis: Application of Serological Markers for the Assessment of Spatial Heterogeneity of *Plasmodium vivax* infections in Papua New Guinea

Undergraduate Student Supervisor, Caitlin Kane

Walter and Eliza Hall Institute of Medical Research

Melbourne, Australia

Feb–Dec 2021

- Designed and supervised a six-month research project in genomics (both lab and analytics) for a Biomedical undergraduate student
- Research project: Amplification of six novel markers for *Plasmodium vivax* genotyping by targeted amplicon sequencing

PhD Student Co-Supervisor, Caitlin Bourke

Walter and Eliza Hall Institute of Medical Research

Melbourne, Australia

Jul 2019–Dec 2023

- Co-supervised a Medical Biology PhD student on *Plasmodium vivax* genomics and bioinformatics
- Thesis: Multi-omic investigations of *Plasmodium vivax*

PhD Student Co-Supervisor, Jason Rosado

Walter and Eliza Hall Institute of Medical Research

Melbourne, Australia

Jul 2019–Jun 2021

- Co-supervised a PhD student on *Plasmodium vivax* genomics (both lab and analytics) and population genetics
- Thesis: Development of novel genotyping tools for *Plasmodium vivax* and the sero-epidemiology of *Plasmodium vivax* in Peru

MSc Student Co-Supervisor, Dionne Argyropoulos

The University of Melbourne

Melbourne, Australia

Feb 2018–Dec 2019

- Co-supervised a Genetics MSc student in genetic epidemiology, population genetics and statistical genetics
- Thesis: The effects of seasonality on the population genetics of the reservoir of asymptomatic *Plasmodium falciparum* in Bongo District, Ghana

Undergraduate Student Supervisor, Angela Hu

The University of Melbourne

Melbourne, Australia

Jul 2016–Nov 2016

- Designed and supervised a six-month research project in statistical genetics and genetic epidemiology for a Computational Biology undergraduate student through the Undergraduate Research Opportunities Program scheme
- Research project: Host genetic factors that influence malaria parasitemia in Ghanaian children

Teaching Experience

I have a strong commitment to mentorship, capacity-building and training, some of the workshops I have (co-)led:

Malaria Molecular Surveillance Study Design workshop co-leader

Imperial College London

Toronto, USA

Nov 2025

- Led a two-day workshop before the ASTMH Annual meeting for researchers, national malaria control program officers and ministry of health representatives on study design for malaria molecular surveillance

Infectious Disease Modelling and Analytics for Pandemic Preparedness short course co-leader

Imperial College London, Universidad San Francisco de Quito, Oxford University

Toronto, USA

Oct 2025

- Led a two-week short course for Latin American researchers and public health officials on epidemic modelling and analytical methods for pandemic preparedness

Malaria Molecular Surveillance Study Design workshop co-leader [↗](#)

New Orleans, USA

Imperial College London

Nov 2024

- Led a two-day workshop before the ASTMH Annual meeting for researchers, national malaria control program officers and ministry of health representatives on study design for malaria molecular surveillance

DRpower *pfhrp2/3* planner workshop leader

Kigali, Rwanda

Multilateral Initiatives for Malaria 8th Pan-African Conference

Apr 2024

- Led a workshop before the MIM Kigali meeting for African researchers and national malaria control program officers on how to use the DRpower *pfhrp2/3* planner

RADISH23 Hackathon Co-organizer

Baltimore, USA

Johns Hopkins Bloomberg School of Public Health

Dec 2023

- Co-organized a four-day hackathon called the Reproducibility, Accessibility, Documentation, and Implementation Software Hackathon in 2023 (RADISH23) to improve usability of data analysis tools for malaria genomics

DRpower *pfhrp2/3* planner software session leader

Dakar, Senegal

Malaria Genomics Convening

Oct 2023

- Led a software session during the Malaria Genomics Convening for African researchers on how to use the DRpower tool

Applied Malaria Molecular Surveillance workshop Co-organizer [↗](#)

Addis Ababa, Ethiopia

Ethiopia Public Health Institute

Aug 2022

- Co-organized a four-day workshop for African researchers on how to use R-based data analysis tools for malaria genomics

Shiny application workshop leader [↗](#)

Virtual

WEHI/Burnet/PNGIMR

March 2022

- Led a four-day workshop on how to use the covidClassifyR Shiny web application, designed for researchers from Papua New Guinea Institute of Medical Research and partner institutions (Materials here [↗](#))

Guest Lecturer

Quito, Ecuador

Universidad San Francisco de Quito

Dec 2021

- Guest lecturer for the Masters of Innovation on COVID-19 epidemiology, variant dynamics and implications for public health in Ecuador

ICEMR Workshop Co-Organizer

Melbourne, Australia

Walter and Eliza Hall Institute

Sep-Dec 2019

- Co-organized an ICEMR-funded hands-on workshop to train six malaria-endemic country scientists on amplicon sequencing (AmpSeq) for *Plasmodium falciparum*

Guest Lecturer

Melbourne, Australia

Nossal Institute for Global Health, University of Melbourne

Oct 2019-Sep 2022

- Guest lecturer for the Nossal Institute for Global Health Masters of Public Health on Malaria and Child Health (POPH90086: Global Child Health)

STEM Workshop Co-Leader

Melbourne, Australia

It Takes a Spark Melbourne Conference 2019

Sep 2019

- Co-led the development of a workshop to teach high-school girls epidemiology and R coding skills to solve an outbreak and become a "disease detective". (Materials here [↗](#))

Guest Lecturer

Quito, Ecuador

Universidad de los Hemisferios

Dec 2017

- Guest lecturer for the Masters of Entrepreneurship on the topic of Social Impact Business Models and The Artisan Project as a Case Study

Laboratory Demonstrator

Melbourne, Australia

The University of Melbourne

Jul 2015-Aug 2017

- Led groups of high school students attending the Victorian Certificate of Education Biology workshops in genetics

Co-Organizer

Quito, Ecuador

ResBaz Ecuador

Feb 2015-Feb 2016

- The Research Bazaar is a worldwide festival promoting the digital literacy emerging at the centre of modern research. I was co-organizer of the first ResBaz in the Americas in 2015 and 2016

Tutor and Teaching Assistant

The Colorado College

Colorado, USA

Sep 2008–May 2012

- Biology Department and Quantitative Reasoning Center, courses included: Introduction to Molecular Genetics, Introduction to Cellular Biology, Advanced Genetics Courses
- Spanish Department, courses included: Beginner, Intermediate and Advanced Spanish and Cultural Context and Written Expression

Leadership, Academic Service and Public Engagement

Manuscript peer-review: Nature Communications (1), Journal of Infectious Diseases (1), PLOS Genetics (1), Malaria Journal (2), International Journal of Parasitology (1), Frontiers in Genetics (2), Infection, Genetics and Evolution (1), Scientific Reports (2), Annals of Global Health (1), Bioinformatics (1), AJTMH (1)

Academic Editor: PLOS Global Public Health (Online)

May 2025–Present

Member, Imperial College London Latin American Staff: Imperial College London (London, UK)

Jul 2024–Jan 2025

Guest Editor: Frontiers in Genetics (Online)

Jan 2024–Present

- Guest editor for the "Advanced Genetic and Genomic Methods and Applications for Malaria Surveillance Special Issue" 

Panelist : Universidad San Francisco de Quito (Quito, Ecuador)

Jan 2021

- Panelist on the panel discussion "Lessons learned from COVID-19 variants" (Lecciones aprendidas con variantes COVID-19) hosted by USFQ and national newspaper El Universo

TV interview : Visionarias EC News Program (Quito, Ecuador)

Dec 2021

- Interview on a national news program on the segment "Omicron - the new variant affecting the globe" (Ómicron - la nueva variante que esta afectando al mundo)

Local Organizing Committee: International Conference on *Plasmodium vivax* research (Melbourne, Australia)

Aug 2021–Mar 2022

President and Co-organizer: R-Ladies Melbourne (Melbourne, Australia)

Aug 2021–Sep 2022

- R-Ladies Melbourne is a non-profit organization aiming to promote gender diversity in the R statistical programming community

Member, Web-based applications working group: Walter and Eliza Hall Institute (Melbourne, Australia)

Jul 2021–Sep 2022

Mentor: CareerTrackers Young Indigenous Women in STEM Academy (Melbourne, Australia)

Dec 2020–Sep 2022

Member, Postdoctoral Association: Walter and Eliza Hall Institute (Melbourne, Australia)

Dec 2020–Sep 2022

Crowdfunding for SARS-CoV-2 serosurveillance in Quito, Ecuador: Collaboration with Ecuadorian Red Cross

(Quito, Ecuador)

Sep 2020–Present

- Led the establishment and provide ongoing support for serosurveillance of SARS-CoV-2 antibodies in blood donors in Ecuador to support COVID-19 epidemiological surveillance. Early in the pandemic, crowdfunding for this initiative raised US\$2,500

Member, Reconciliation Committee: Walter and Eliza Hall Institute (Melbourne, Australia)

Jun 2019–Sep 2022

- The RC aims to contribute towards reconciliation by working towards 'closing the gap' in life expectancy, disease incidence, and mortality for Aboriginal and Torres Strait Islander Australians

Committee Chair, Annual Conference: Australian Centre for Excellence in Malaria Elimination (Australia)

Mar–Oct 2019

Social Impact Entrepreneur: Melbourne Accelerator Program (Melbourne, Australia)

Sep 2016–Feb 2017

- Selected to represent The Artisan Project (at the time I was CEO) in the MAP Social Velocity six-month competitive entry program for early stage social impact start-ups

Entrepreneur: European Innovation Academy (Nice, France)

Jul 2016

- Selected as one of four students to represent University of Melbourne at EIA, the world's largest extreme accelerator program for tech and digital innovations

Professional Memberships

2022–Present Member, Plasmodium Genomic Epidemiology (PlasmoGenEpi) Network

2022–Present Member, Applied Malaria Modeling Network

2019–2022 Research Fellow, Australian Centre of Research Excellence in Malaria Elimination

2019–2022 Research Fellow, Australian Centre for Control & Elimination of Neglected Tropical Diseases

2015–2017 Secretary & President, Ecuadorian Research & Entrepreneurship Network

2015–2022 Member, Australian Society for Parasitology

2012–Present Member, American Society for Tropical Medicine & Hygiene

Additional training and professional development

- 2024 **International Workshop on Epidemic Preparedness for Infectious Diseases in the Brazilian Amazon Region** (Imperial College London and Tropical Medicine Foundation)
- 2023 **Short course: Modelling the Epidemiology & Control of Infectious Diseases** (Imperial College London)
- 2021 **Short course: Multiple imputation** (Victorian Centre for Biostatistics)
- 2021 **RStudio Diversity Scholar Workshops** (RStudio)
- 2018 **Certification "Teaching Skills for Graduate Researchers"** (University of Melbourne)
- 2017 **Winter School in Mathematical Biology** (Institute for Molecular Science, The University of Queensland)
- 2015 **Short course: Spatial mapping and GIS skills** (University of Melbourne)
- 2015 **UOM subjects: Epidemiology 1, Biostatistics 1, Linear & Logistic Regression** (University of Melbourne)

Skills

Laboratory	Analytical	Programming	Software/Tools
DNA extraction	Epidemiological database curation/management	R (advanced)	Git/Github
PCR/qPCR	Population genetics	Rstudio	REDCap
Serology Luminex assays	Bioinformatics	Rmarkdown/Quarto	LaTeX
Illumina amplicon sequencing	Statistical modelling and Bayesian inference	SLURM/high-performance computing	CSS
PacBio long-read sequencing	Mathematical/epidemiological modelling	Python	HTML
Whole-genome sequencing	Machine learning approaches	Command line	Reproducible research

Awards and Funding

Junior Leader Fellowship

La Caixa Foundation

Jun 2025

- Awarded €320,100 as a Junior Leader Fellow supporting researchers with an exceptional track record and strong leadership potential, with the capacity to lead their own research group in the future

Seed Grant (Chief Investigator)

MRC-GIDA, Imperial College London & Leverhulme Centre for Demographic Science, Oxford University

Jan 2025

- Awarded £55,000 as CI to establish a collaborative training partnership, Infectious Diseases Modelling and Analytics for Pandemic Preparedness in Latin America (IDMAPP-LATAM) to build key epidemiological modelling analytical capacities and support regional operational research and pandemic preparedness efforts

Seed Grant (Associate Investigator)

Australian Centre for Research Excellence in Malaria Elimination

Dec 2021

- Awarded A\$15,380 as AI for the collaborative project: Artificial Intelligence-based drug resistance screening of malaria parasites using Read Until
- Awarded A\$14,061 as AI for the collaborative project: Methodological comparison of *P. vivax* genotyping approaches for molecular surveillance within the ACREME network

Research Grant (Associate Investigator)

Australian Centre for Research Excellence in Malaria Elimination

Sep 2021

- Awarded A\$100,000 as AI for the collaborative project: Developing a MinION long-read deep amplicon sequencing assay for host variants relevant to 8-aminoquinolines administration

COVID-19 Digital Grant (Chief Investigator)

Australian Academy of Science & Dept of Industry, Science, Energy & Resources

Apr 2021

- Awarded A\$10,000 as CI for the collaborative project: Fit-for-purpose analytical tools to support COVID-19 sero-surveillance in Papua New Guinea [↗](#)

Professional Development Award

Walter and Eliza Hall Institute of Medical Research

Jan 2021

- Awarded to attend the Victorian Centre for Biostatistics Summer School

RStudio Diversity Scholar

RStudio

Jan 2021

- Selected as a Diversity Scholar at the rstudio::global(2021) conference to attend the DS post-conference workshops

Ideas Jam Seed Grant (Associate Investigator)

Burnet Institute

Oct 2020

- Awarded A\$10,000 as AI for the project: Superstars in STEM High School Program

Seed Grant (Associate Investigator)

Australian Centre for Control & Elimination of Neglected Tropical Diseases

Oct 2020

- Awarded a total of A\$100,000 (Oct 2020 and Jul 2021) as AI for the collaborative project: Burden of neglected tropical diseases on the north coast of Papua New Guinea

Seed Grant (Associate Investigator)

Australian Centre for Research Excellence in Malaria Elimination

Sep 2020

- Awarded A\$14,400 as AI for the collaborative project: Validating molecular and serological tools for detecting hidden reservoirs of *Plasmodium* infections in Papua New Guinea

Kellaway Excellence Award in Education

Walter and Eliza Hall Institute of Medical Research

Dec 2019

- Awarded to recognize the significant contribution to improve female representation in statistical bioinformatics through joint leadership of R-Ladies Melbourne, the R programming community for women in Australia

Seed Grant (Chief Investigator)

Australian Centre for Research Excellence in Malaria Elimination

Jul 2019

- Awarded A\$15,000 as CI for the collaborative project: Building capacity for an innovative amplicon deep sequencing tool to genotype *Plasmodium* infections for improved malaria surveillance

Three Minute Thesis (3MT) Award 1st Place

The University of Melbourne

Jun 2018

- Awarded to the best 3MT presentation by a Faculty of Science postgraduate student

Travel Bursary

Wellcome Genome Campus

May 2018

- Awarded to present my PhD findings at the International Genomic Epidemiology of Malaria Conference in UK

Science Abroad Traveling Scholarship

The University of Melbourne

Oct 2017

- Awarded to present my PhD findings at the 2017 American Society for Tropical Medicine and Hygiene conference in USA

JD Smyth Postgraduate Student Travel Award

Australian Society for Parasitology

Sep 2017

- Awarded to undertake a researcher exchange at the Center for Research on Health in Latin America in Ecuador [↗](#)

Dame Margaret Blackwood Soroptimist Scholarship

The University of Melbourne

May 2017

- Awarded to an outstanding female student undertaking research in genetics & who is engaged in the world beyond academia. Recognized as a Dame Margaret Blackwood Scholar [↗](#)

European Innovation Academy Travel Scholarship

The University of Melbourne

Jul 2016

- Selected as one of four students to represent The University of Melbourne at the EIA and awarded with a travel scholarship to attend the three-week program on tech, entrepreneurship and innovation in Nice, France

Postgraduate Student Travel Award

Bio21 Institute of Molecular Science & Biotechnology

Sep 2014

- Awarded to present my PhD findings at the 2014 American Society for Tropical Medicine and Hygiene conference in USA

Publications

Peer-reviewed manuscripts and pre-prints

2026

Bansal, A., Olechnowicz, S.W.Z., Kiernan-Walker, N., Cumming, J.,, **COVID-PROFILE consortium**, Cox, R.J., Mueller, I., Bowden, R., Eriksson, E.M. (2026) *Divergent inflammatory and neurology-related protein levels in long COVID following primary and breakthrough SARS-CoV-2 infections*. Communications Medicine. <https://doi.org/10.1038/s43856-026-01541-6>

Ruybal-Pesántez, S., Amaya-Romero, J., Bérubé, S., Brazeau, N. F., Diop, M. F., Hathaway, N., Hendry, J., McCann, K., Murie, K., Murphy, M., Niaré, K., Phelan, J., Schaffner, S. F., Simkin, A., Taylor, A. R., Greenhouse, B., Wesolowski, A., Verity, R. (2026, *accepted*)

in AJTMH) Towards an open analysis ecosystem for Plasmodium genomic epidemiology. medRxiv. <https://doi.org/10.1101/2025.04.01.25325032v1>

Rosado, J., Han, J., Obadia, T., Munro, J., Traore, Z., Schoffer, K., Brewster, J., Bourke, C., Vinetz, J., White, M., Bahlo, M., Gamboa, D., Mueller, I., **Ruybal-Pesántez, S.** (2026, accepted in iScience). Understanding Plasmodium vivax recurrent infections using an amplicon deep sequencing assay, PvAmpSeq, identity-by-descent and model-based classification. medRxiv. <https://doi.org/10.1101/2025.05.26.25327775>

Naidoo, T., Morgenstern, C., Doohan, P., Earl, R., ..., **Pathogen Epidemiology Review Group**, van Elsland, S., Cori, A., McCabe, R., Bhatia, S. (2026, under review) A systematic review of Nipah virus disease epidemiological parameters, outbreaks and mathematical models medRxiv. <https://doi.org/10.64898/2026.03.19.26348815>

2025

Sheppard, R.J., Charles, G.D., Ciavarella, C., Schmidt, N., **Ruybal-Pesántez, S.**, Cuomo-Dannenburg, G., Brewer, T.R., White, M.T., Winskill, P. (2025, under review). Integrating parallel Plasmodium falciparum and Plasmodium vivax malaria models in a unified framework to capture co-endemic prevalence patterns.

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- This work was featured on the cover of the February issue 

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- This work was featured on the Herminthology #WomenBehindTheWork initiative 

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- This work was chosen by the editors to be featured in the Molecular Ecology blog 

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- This work was chosen by the editors to be featured with an accompanying Perspectives piece 

2020

Narh, C. A., Ghansah, A., Duffy, M. F., **Ruybal-Pesántez, S.**, Onwona, C. O., Oduro, A. R., Koram, K. A., Day, K. P.* , & Tiedje, K. E.* (2020). *Evolution of antimalarial drug resistance markers in the reservoir of Plasmodium falciparum infections in the Upper East Region of Ghana*. The Journal of Infectious Diseases. <https://doi.org/10.1093/infdis/jiaa286>

2019

Pilosof, S., He, Q., Tiedje, K. E., **Ruybal-Pesántez, S.**, Day, K. P., & Pascual, M. (2019). *Competition for hosts modulates vast antigenic diversity to generate persistent strain structure in Plasmodium falciparum*. PLOS Biology, 17(6), e3000336. <https://doi.org/10.1371/journal.pbio.3000336>

2018

He, Q., Pilosof, S., Tiedje, K. E., **Ruybal-Pesántez, S.**, Artzy-Randrup, Y., Baskerville, E. B., Day, K. P., & Pascual, M. (2018). *Networks of genetic similarity reveal non-neutral processes shape strain structure in Plasmodium falciparum*. Nature Communications, 9(1), 1817. <https://doi.org/10.1038/s41467-018-04219-3>

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Ruybal-Pesántez, S., Tiedje, K. E., Rorick, M. M., Amenga-Etego, L., Ghansah, A., Oduro, A. R., Koram, K. A., & Day, K. P. (2017). *Lack of Geospatial Population Structure Yet Significant Linkage Disequilibrium in the Reservoir of Plasmodium falciparum in Bongo District, Ghana*. The American Journal of Tropical Medicine and Hygiene, 97(4), 1180--1189. <https://doi.org/10.4269/ajtmh.17-0119>

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Book chapters

Dori, A., Farquhar, R., Kelebi, T., Waipeli, E., Kerry, Z., **Ruybal-Pesántez, S.**, Timbi, D., McEwen, S., Makita, L., Laman, M., Robinson, L.J. (2025). *Partnership-based approach to infectious disease research in Papua New Guinea*. Transforming Global Health Partnerships: Critical Reflections and Visions of Equity at the Research-Practice Interface, edited by Stewart-Ibarra, A. and LaBeaud, A.D., Springer 1st Edition. <https://doi.org/10.1007/978-3-031-53793-6>

Technical reports

Winskill, P., Haile, L., **Ruybal-Pesántez, S.**, Simmons, O., Topazian, H., Okell, L. (2025). *Rapid response modelled estimates of the effect of the US global aid freeze on President's Malaria Initiative impact in sub-Saharan Africa*. Technical report, Imperial College London. <https://hdl.handle.net/10044/1/118190>

Digital tools and software

For other non-traditional academic contributions, I have also developed several **R Shiny** web applications, packages, and dashboards. Check out my GitHub [🔗](#)

plasmodiff [🔗](#)

Shazia Ruybal-Pesántez

In development

- This R package provides a diffusion network-based framework for reconstructing malaria transmission networks from genomic and epidemiological data

malariafwd [🔗](#)

Shazia Ruybal-Pesántez

In development

- This R package provides a simulation-based framework for Plasmodium genetic epidemiology

varcodetools [🔗](#)

Shazia Ruybal-Pesántez

In development

- This R package provides an analysis toolkit for analysis of DBLalpha varcoding amplicon sequencing data

epireview [🔗](#)

Pathogen Epidemiology Review Group

In development

- This R package compiles epidemiological parameters from systematic reviews of the WHO priority pathogens with pandemic potential

SIMPLEGEN [🔗](#)

Shazia Ruybal-Pesántez and Bob Verity

In development

- This R package simulates malaria genetic epidemiology to explore the “power” of genomic surveillance sampling designs.

PGEcore [🔗](#)

PlasmoGenEpi Network

In development

- This is a core set of scripts to perform common downstream analysis functionalities for Plasmodium genomic analysis

PGEforge [🔗](#)

Shazia Ruybal-Pesántez, PlasmoGenEpi Network

July 2025

- This is a central, open repository for current and future *Plasmodium* genetic data and analysis workflows.

DRpower pfhrp2/3 planner [🔗](#)

Shazia Ruybal-Pesántez and Bob Verity

Nov 2023

- This Shiny web application simplifies the design of studies that monitor rapid diagnostic tests effectiveness for malaria diagnosis in the field and uses the DRpower R package in the back-end.

- This R package is designed to assist with the development of *Plasmodium falciparum* pfhrp2/3 gene deletion studies.

CovidClassifyR

Shazia Ruybal-Pesántez

Sep 2021

- This Shiny web app was developed to support COVID-19 serosurveillance in Papua New Guinea enabling classification of unknown samples as recently exposed to SARS-CoV-2 by streamlining QC, processing and interpretation of raw data generated from a validated COVID-19 serological assay (Mazhari et al 2020). Funding: COVID-19 Digital Grant, media release 

COVID-19 VaccinationScore

Raúl Fernández, Shazia Ruybal-Pesántez, Esteban Ortíz

Feb 2021

- This Shiny web app was developed during the initial COVID-19 vaccine roll-out in Ecuador to help individuals understand their "priority status" to receive their vaccine. An algorithm was applied to calculate a priority score based on socioeconomic status, occupation, exposure, risk behavior, comorbidities, etc. Newspaper article (in Spanish) 

Serosurveillance of COVID-19 in Ecuadorian blood donors dashboard (not open-source)

Shazia Ruybal-Pesántez

Jun 2020

- This R flexdashboard presented seroprevalence trends from monthly screening of blood donor samples in collaboration with the Ecuadorian Red Cross National Blood Bank as part of the emergency COVID-19 response (active early in the pandemic, June 2020-Dec 2020).

ICEMR weekly dashboard (not open-source)

Shazia Ruybal-Pesántez

Mar 2020

- This R flexdashboard was developed to support ICEMR field teams in Madang, PNG (active during cohort study Mar 2020-Sep 2021) with weekly operational data (e.g. follow-up rates) and preliminary epi trends (e.g. RDT positivity, prevalence of fever).

Selected presentations

I have participated in oral and poster presentations at >20 conferences (6 travel awards).

- 2026 **Genetic epidemiology of *P. vivax* resurgence on the Thai-Myanmar border**, Invited to speak at the Gates Foundation Vivax Convening in Bangkok, Thailand
- 2025 **Genomic surveillance and pandemic preparedness**, Keynote speaker at the "Infectious Disease Modelling and Analytics for Pandemic Preparedness in Latin America (IDMAPP-LATAM)" short course in Quito, Ecuador
- 2024 **Towards an open, function analysis ecosystem: PGEforge and the way forward for data and analysis standards**, Invited to speak at the "Harmonising data standards and analysis resources for public health" session at the Genomic Epidemiology of Malaria conference in Hinxton, UK
- 2024 **SIMPLEGEN: a modelling approach (de-)coupling epidemiology and genomics to inform malaria surveillance**, Invited to speak at IRD Paris seminar series
- 2024 **A gentle introduction to malaria genomic modelling**, Invited as a keynote speaker at the Applied Malaria Modelling Network Annual Meeting in Kigali, Rwanda
- 2023 **Using genomic tools to understand malaria transmission patterns and support surveillance: using reactive case detection data to map parasite connectivity**, Invited to speak at the Zanzibar Imported Malaria (ZIM) Study meeting with the Zanzibar Malaria Elimination Program in Zanzibar, Tanzania
- 2022 **Establishing multi-pathogen sero-surveillance to inform disease control interventions in Papua New Guinea**, Invited speaker at the "Laboratory Implementation of Multiplex Serologic Methods: Multi-Pathogen Integrated Surveillance to Inform Public Health Decision-Making in Diverse Geographic Settings" symposium at ASTMH in Seattle, USA
- 2021 **Understanding the factors underlying malaria resurgence in East Sepik, Papua New Guinea: a preliminary analysis**, Invited to speak at the Australian Centre for Excellence in Malaria Elimination webinar series. See recording 
- 2019 **Resurgence of *Plasmodium falciparum* after an outbreak in Ecuador is driven by local transmission and parasite evolution via recombination**, Oral presentation at the Malaria in Melbourne conference, Melbourne, Australia